

Shifting Tornado Hazard in the United States and Emerging Insurance Exposure

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1 Abstract

This capstone project examines whether tornado hazard in the United States is shifting geographically from the traditional Great Plains "Tornado Alley" region toward the Southeast's "Dixie Alley" region, and whether this shift is associated with increased insurance exposure. Building upon prior research by Vittorio Gensini and Harold Brooks (2018), this project integrates tornado hazard data with population and housing information to evaluate whether emerging high-risk counties also represent a greater financial risk to insurers. The findings indicate a growing concentration of tornado hazard within the Dixie Alley region and include meaningful insurance implications for risk assessment, underwriting strategies, and long-term disaster planning.

2 Introduction

There are two major regions in the United States that are commonly associated with elevated tornado activity: the Southeast's region of Dixie Alley, which includes Arkansas, Louisiana, Mississippi, Alabama, Tennessee and Georgia, and the Great Plains region of "Tornado Alley", which includes Texas, Oklahoma, Kansas and Nebraska. Recent research indicates that tornado activity may be shifting geographically while increasing in frequency in Dixie Alley and decreasing in frequency in Tornado Alley (Gensini and Brooks, 2018).

This capstone project investigates whether tornado hazard in the United States is shifting towards Dixie Alley over time, and whether counties indicating elevated tornado hazard also pose a higher risk of insurance exposure. Insurance exposure is defined as the potential financial risk to insurers, approximated using county-level population, housing units and median home value.

Understanding this potential shift is important from an insurance perspective. If tornado activity is becoming more frequent and severe in areas with higher housing and population density, like Dixie Alley, insurance companies may be faced with an increased exposure and loss potential (Memphis Weather, 2025). Though the idea of this shift in tornado history has been previously explored, this project extends upon this while incorporating an insurance lens with a predictive modeling framework. It will ultimately examine whether elevated tornado hazard is shifting towards Dixie Alley and occurring in counties with greater insurance exposure.

2.1 Research Question

1. Is tornado hazard in the United States shifting geographically from Tornado Alley to Dixie Alley over time?
2. Are counties with elevated tornado hazard associated with higher insurance exposure?

2.2 Hypothesis and Prediction

The hypothesis is that tornado hazard is shifting from the Tornado Alley region toward Dixie Alley due to increasing tornado frequency and severity in the Southeast over time. It is predicted that counties located in Dixie Alley will be more likely classified as high tornado hazard in the following year compared to Tornado Alley and will exhibit a higher present-day insurance exposure.

3 Data

3.1 Data Acquisition

The historical tornado data was obtained from the National Oceanic and Atmospheric Administration (NOAA) Severe Weather Database file (1950-2024) and downloaded as a CSV file. It includes detailed tornado records across the United States including the event date (year, month and day), location, tornado intensity (F-Scale or EF-Scale) and estimated property loss. County-level exposure data was obtained using the tidycensus package in R and the American Community Survey (ACS) 5-year estimates for the variable 2024 for Estimate Total Population (B01003_001), Estimate Total Housing Units (B25001_001) and Estimate Median Value (Dollars) (B25077_001). These ACS datasets were selected due to their relevance with insurance data exposure to serve as insurance proxies detailing the estimated population, total housing units and median home value which are all important variables for the insurance stakeholder when determining potential policies in a given location.

The predictor variables include estimated population, housing units, median home value and tornado characteristics (width and length). The response variable for this analysis, High Tornado Hazard Next Year, is a binary indicator representing whether a county was predicted to be classified as a high tornado hazard in the following year.

The Data Dictionary for all variables is found in the Appendix.

3.2 Data Limitations

NOAA data included inconsistencies over time, including inconsistent property loss reporting prior to and after 1996 and missing, or unknown, intensity observations. Additionally, the ACS data is used as insurance exposure, however, may not fully capture insurance exposure as well as insurance claims.

3.3 Data Cleaning

All data cleaning and exploration was conducted in RStudio (version 2024.12.1). The datasets were merged using the common FIPS identifier to combine datasets with historical tornado data and present-day exposure variables. The tornado dataset was cleaned by creating a new column for the complete five-digit FIPS code, combining the state FIPS (stf) and county FIPS (fl-f4) columns, and restoring any leading zeroes omitted. The originating county (fl) was retained to simplify analysis at the county-year level.

Columns that were not needed for this project were removed, including injuries. The dataset was filtered by year to only retain observations between 1979-2024. This aligns with the original research data from Gensini and Brooks. The tornado events were then aggregated to a county-year level so that each observation represented one county in one year. This allows alignment with predicting which counties will flag as high hazard the following year using the target variable.

Missing or invalid variables were also addressed. The variable mag, which represents the F/EF-Scale 0-5, also included observations for -9 or unknown. These were replaced to indicate as missing or NA so they do not distort averages or max values of other engineered features. These NA values will contribute to inevitable missingness, however, the decision was necessary to ensure accuracy with the dataset. The values were then standardized across the F-Scale and EF-Scale to a common scale of 0-5.

Prior to 1996, the tornado property loss was recorded by NOAA using categorization by dollar amount where codes represent damage ranges: 0 = unknown or blank, 1 < \$50, 2 = \$50-\$400, 3 = \$500-\$5,000, etc. Anything from 1996 onward includes the tornado property damage in millions of dollars. As 0 represents unknown rather than a true zero, they were also replaced

as missing (NA) to avoid distorting averages or max values. These will also contribute to missingness. To ensure consistency, pre-1996 categorical values were converted into approximate midpoint values in millions of dollars and post-1996 values remained their original value. This allowed for a consistent reporting of property loss for aggregation and modeling across the full timeframe.

3.4 Feature Engineering

Regional definitions were created defining the states within Tornado Alley and Dixie Alley using state FIPS codes allowing for feature creation classifying the regions. Counties were classified into three different regions based on their state:

- **Tornado Alley:** Texas, Oklahoma, Kansas and Nebraska
- **Dixie Alley:** Louisiana, Arkansas, Tennessee, Georgia, Alabama and Mississippi
- **Other:** Any other state

These categorical features for Region Category were encoded as a factor variable to assess the geographical differences in tornado hazard. This will allow for use in modeling.

The dataset was then aggregated from tornado-level observations to a county-year level which aligns with the goal of predictive modeling. Each observation represents one county per year. For each county-year observation, the following features were engineered:

- Tornado Count per County-Year
- Average Tornado Intensity per County-Year
- Maximum Tornado Intensity per County-Year
- Strong Tornado Count per County-Year (F/EF 3 or greater)
- Total Property Loss per County-Year
- Average Property Loss per County-Year
- Average Tornado Length per County-Year
- Average Tornado Width per County-Year

Additional variables included:

- Property Loss per Tornado

- Strong Tornado Fraction

To account for differences in population and housing density, normalized variables were created:

- Tornadoes per 100,000 Population
- Tornadoes per 10,000 Housing Units

The ACS Survey insurance exposure data (population estimate, housing units and median home value) was then merged into the tornado dataset at a county-level. Highly skewed variables, including property loss, population and housing units, were log-transformed to improve model performance.

3.5 Target Variable

The target variable, High Tornado Hazard Next Year, was intentionally created with variables different from the predictor variables to avoid data leakage during modeling. The composite hazard score included Tornado Count per County Year, Average Intensity County, Max Intensity County, Strong Tornado Count, and (log-transformed) Property Loss. Z-Scores were used on all variables to allow for standardization and ensure comparability across differing scales.

A threshold of 65% was selected to classify the high-risk observations. This was adjusted from the original threshold of 75% to allow for better performance to improve recall and reduce number of missed high-risk observations. This helped address class imbalance and improve the model's ability to accurately identify high-risk counties. Although this may introduce bias by increasing false positives, this is a necessary tradeoff as the most important goal for stakeholders is to accurately identify high-risk counties. Any composite scores at or above the 65th percentile will be flagged as 1 = high hazard next year. Anything under this threshold will be labeled as 0 =

not high hazard next year. This was also shifted forward one year with a lead function allowing the target variable to look at future hazard conditions.

3.6 Missing Data

Missing data values were retained as NA during preprocessing and removed prior to modeling (na.omit) to ensure it did not skew averages or max values within the dataset. Tornado intensity was updated for any -9 values indicating unknown to NA. This assumes missingness is random, however, it may also introduce bias.

3.7 Data Exploration

Univariate, bivariate and multivariate analysis was conducted to understand patterns and relationships in the data prior to predictive modeling. Bivariate analysis included examining year and tornado count trends over the years (1979-2024). Additionally, a bivariate jitter plot was created to examine distribution between region and normalized tornado frequency per 100,000 population, as well as per 10,000 housing units allowing for the identification of outliers. Multivariate analysis included examination of regional trends with the number of tornadoes over the years (1979-2024) within Tornado Alley and Dixie Alley, as well as strong tornado counts by year (F/EF3 and greater).

Normalized variables were created to account for differences in population and housing exposure. This allows for better comparison of tornado frequency across regions. Other feature engineering included regional classification, normalized tornado frequency and exposure-relevant measures for analysis and modeling.

County-year is the selected unit of analysis for the tornado hazard in this project due to the spatial and temporal component. It's valuable for insurance stakeholders to not only understand

the counties with high tornado hazard, but also how that hazard is changing over time. Using county-year observations allows for a deeper analysis into which counties more frequently flag as high hazard, which are becoming more hazardous over time, and ultimately, whether these patterns support a shift towards Dixie Alley potentially representing a higher exposure for insurance companies.

Review of continuous variables was conducted including mean, median, standard deviation, min, max and skewness was performed as well as categorical variables for county and percent. From the continuous variable table, it shows significant right-skewing for population, housing unit estimates and property loss. This shows that most counties have a small amount of property damage due to tornado events while others reported large losses, which will be important for accurate prediction of future hazard counties. Tornado frequency is low but extremely variable, meaning most county-years are experiencing around one tornado but some also show high activity.

Log transformations were applied to variables that were highly right-skewed, including population, housing and property loss and are included in the Continuous Variables summary table. Population skewness decreased from over 11 to 0.318 after transformation. This was extremely important to perform for logistic regression modeling. XGBoost and random forest do not require these transformations so both transformed and untransformed variables will be considered depending on the predictive modeling method.

Variable (per County Year)	Mean	Median	Standard Deviation	Min	Max	Skewness
Population Estimate	140,089	30,986	450,075	33	9,808,667	11.27
Housing Unit Estimate	59,799	14,386	179,507	33	3,650,466	10.22

Median Home Value Estimate	\$208,956	\$176,100	\$114,000	\$48,600	\$1,633,900	2.55
Tornado Count	1.67	1.0	1.31	1	43	4.80
Average Tornado Intensity	0.65	0.5	0.76	0	5	1.30
Max Tornado Intensity	0.8	1	0.90	0	5	1.13
Total Property Loss	\$553,728	\$0.26	\$19,199,710	\$0.0001	\$1,960,040,000	80.17
Average Property Loss	\$329,558	\$0.13	\$11,906,450	\$0.0001	\$1,504,000,000	104.41
Strong Tornado Count (F/EF 3 or greater)	0.054	0	0.25	0	4	5.17
Property Loss per Tornado	\$296,793	\$0.14	\$11,716,420	\$0.0001	\$1,504,000,000	109.13
Tornadoes per 100,000 Population	14.42	4.46	56.80	0.010	6,060.61	51.86
Tornadoes per 10,000 Housing Units	2.76	0.95	7.82	0.0027	606.06	24.81
Average Tornado Length (yards)	3.18	1.08	5.70	0.00	124	5.17
Average Tornado Width (yards)	116.5	50	187	0.00	3,330	4.54
Property Loss (Log Transformed)	11	1.26	4.45 (log scale)	1	2,000,000,000	1.73
Population (Log Transformed)	35,000	31,000	1.58 (log scale)	34	9,800,000	0.32
Housing Unit (Log Transformed)	16,200	14,400	1.52 (log scale)	34	3,650,000	0.37

Table 1: Chart of Continuous Variables summary table
Log-Transformed variables are in original units for readability, however, standard deviation are log scale

Variable	Category	Count	Percent
region_category	Tornado Alley	7,843	24.66%
region_category	Dixie Alley	6,395	20.11%
region_category	Other	17,566	55.23%
high_tornado_hazard	0 = Not High Hazard	23,853	75%

high_tornado_hazard	1 = High Hazard	7,951	25%
high_tornado_hazard_next_year	0 = Not High Hazard	21,617	67.97%
high_tornado_hazard_next_year	1 = High Hazard	7,177	22.57%
high_tornado_hazard_next_year	NA	3,010	9.46%

Table 2: Chart of Categorical Variables summary table

Table 2 shows that approximately 25% of the county-year observations are High Tornado Hazard observations and approximately 75% were not high hazard. Approximately 22% of county-year observations are indicated as High Tornado Hazard next year. More than 50% of county-year observations occurred outside of Tornado and Dixie Alley with approximately 25% occurring in Tornado Alley and 20% in Dixie Alley. The NA value for High Tornado Hazard Next Year looking one year into the future for the last observed year in each county, there is no future data available.

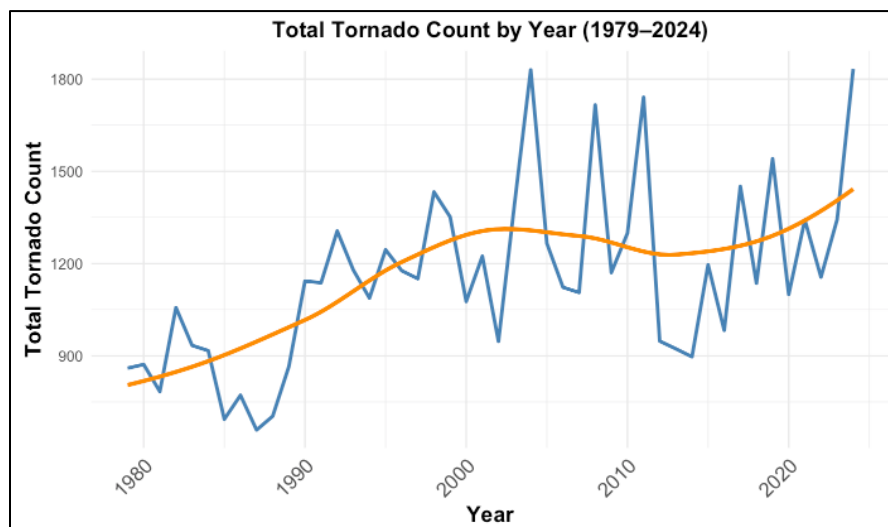


Figure 1: Annual tornado counts in the United States aggregated by year

Figure 1 examines the annual tornado count in the United States between 1979-2024. There is significant variability from year to year with the frequency of tornadoes. This

visualization suggests a possible increase in tornado frequency in recent years, particularly from 2015 onward. However, variability and fluctuation of tornado frequency over time is expected.

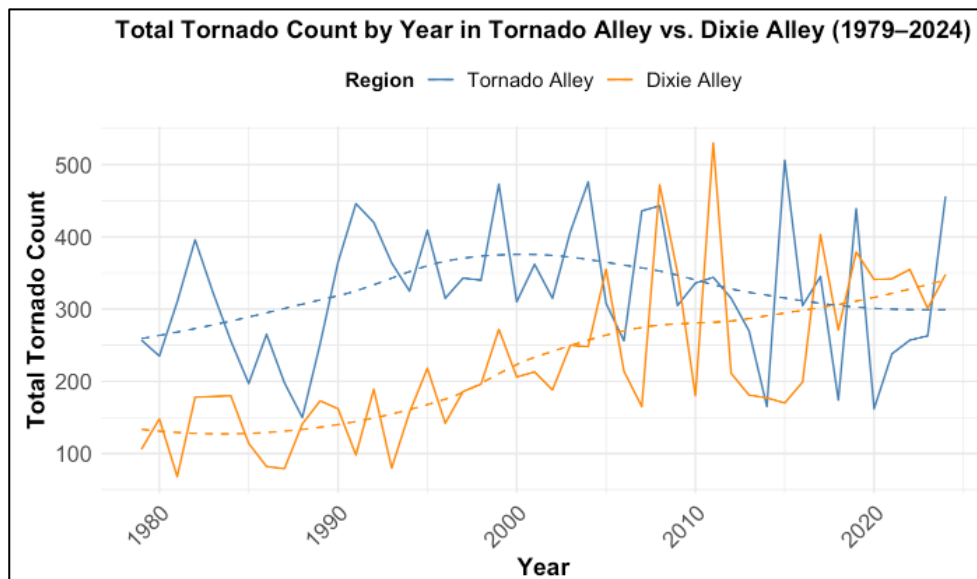


Figure 2: Annual tornado counts by year in Tornado Alley vs. Dixie Alley

Figure 2 shows the annual tornado counts over time in both Tornado Alley and Dixie Alley between 1979-2024. The trend lines suggest that tornado counts in Tornado Alley may have remained stable or slightly declined in recent years, while tornado counts in Dixie Alley appear to have increased in frequency, particularly from 2000 onward. This pattern is consistent with the project hypothesis that greater tornado hazard events may be shifting toward Dixie Alley over time.

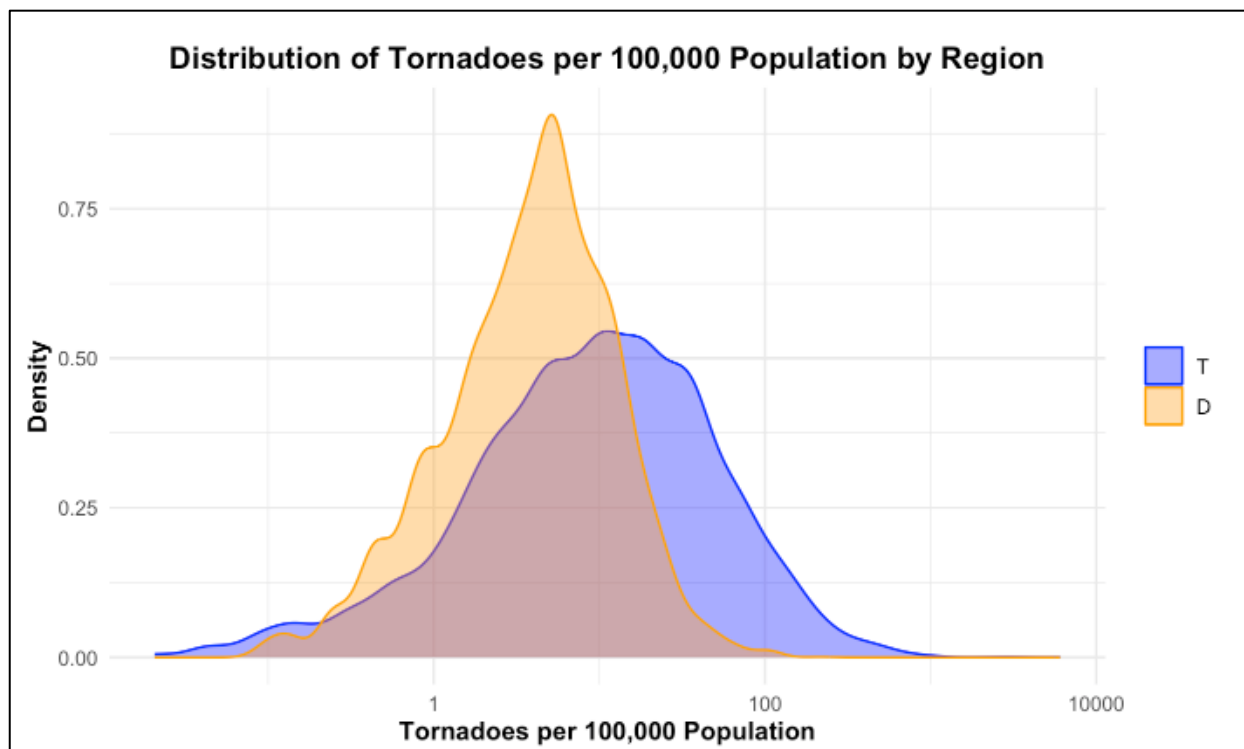


Figure 3: Distribution comparison between Tornado Alley vs. Dixie Alley

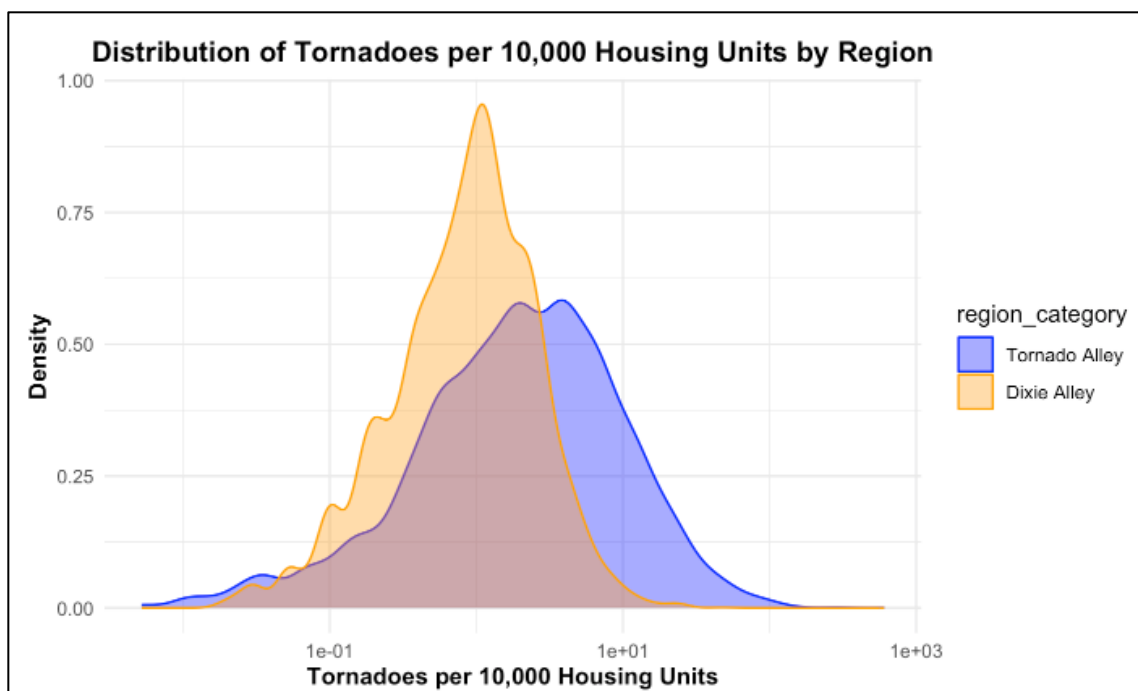


Figure 4: Distribution comparison between Tornado Alley vs. Dixie Alley

Figure 3 shows the distribution comparison of normalized tornado frequency per 100,000 population across the county-year observations in Tornado Alley and Dixie Alley. Figure 4 shows the distribution comparison of tornado frequency per 10,000 housing units across the regions. Both resulted in right-skewed data within both regions indicating that most county-years experience low normalized tornado frequency in addition to a few extreme values. In both graphs, the extreme values occur more frequently within Tornado Alley suggesting tornado occurrences in counties with lower population and lower housing units. Alternatively, Dixie Alley appears to be more consistently concentrated. From an insurance exposure, this could support the hypothesis that the higher hazard is shifting towards Dixie Alley as the risk appears to be more consistent within this location translating to higher risk exposure in comparison to more extreme but less frequency risk in Tornado Alley.

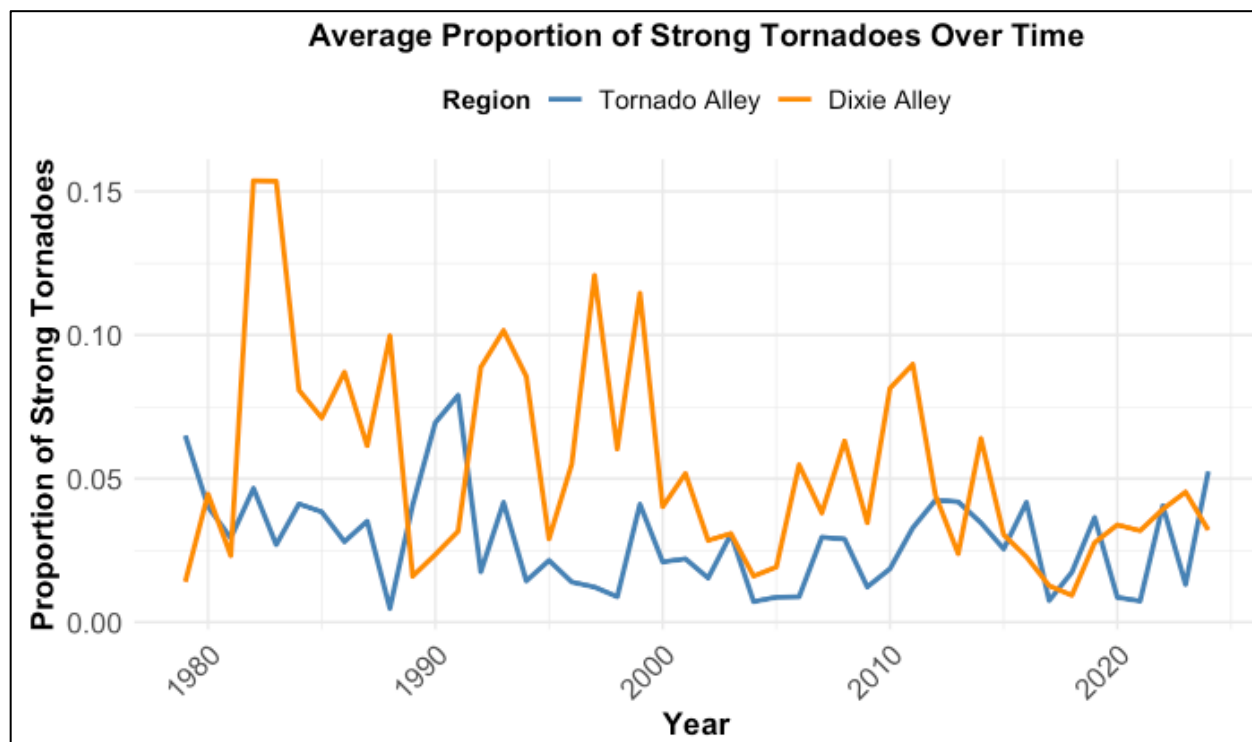


Figure 5: Average proportion of strong tornadoes (F/EF3 or greater) over time

Figure 5 shows the average proportion of strong tornadoes (F/EF3 or greater) over time. This looks at the relative total tornado activity over time detailing the severity of tornado activity. From this graph, Dixie Alley appears to have a consistently increased proportion of strong tornadoes over time compared to Tornado Alley. This poses a risk with insurance exposure with chance of more extreme tornadoes being higher in this area.

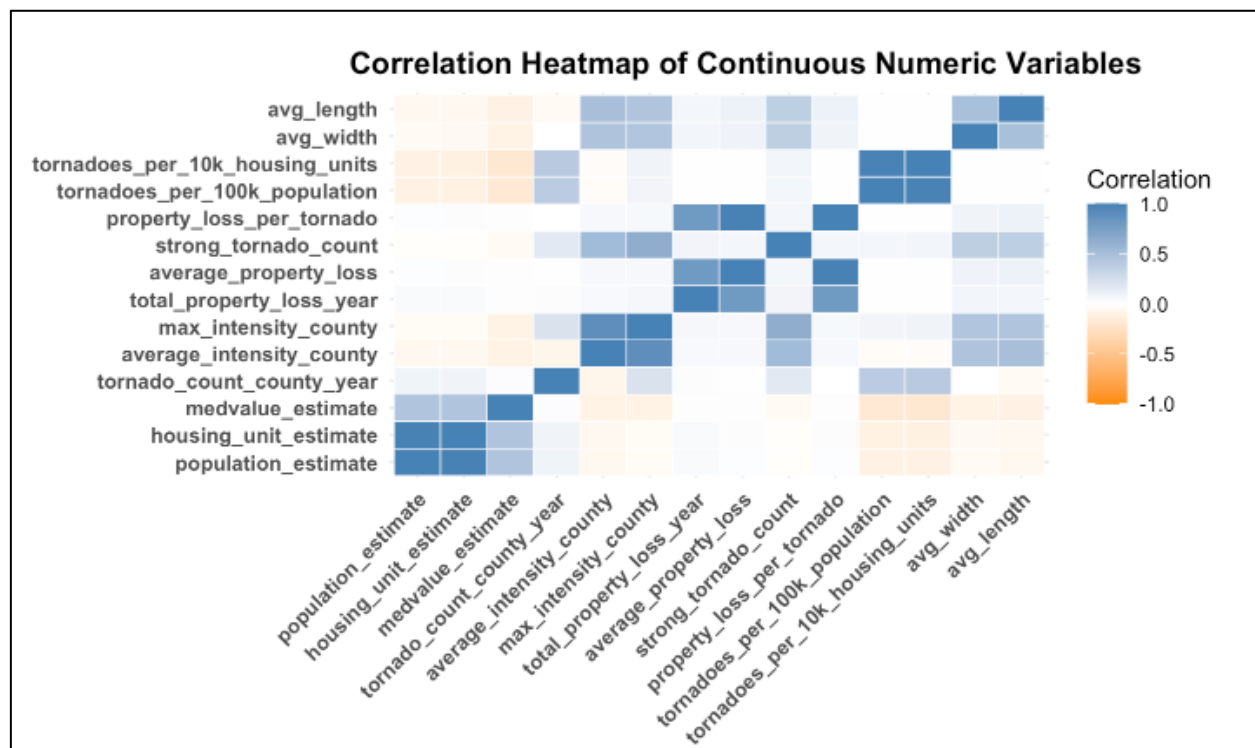


Figure 6: A correlation matrix of continuous numeric variables included in the analysis

Figure 6 details a correlation matrix of the continuous numeric variables that are included in the project analysis. There are strong positive relationships observed, especially between population estimate and housing unit estimate, average and maximum tornado intensity and the normalized tornado rate measures (tornadoes per 100,000 population and per 10,000 housing units). Not all variables are used as predictors as some are used solely for target variable creation. The target variable was created using tornado count per county year, average intensity per

county, max intensity per county, strong tornado count (F/EF 3 or greater) and property loss (logged value). Predictors for the project include average length, average width, estimated population, estimated housing units, estimated median home value and region category. Population and housing show high multicollinearity but are both valuable to this project as they represent different types of insurance exposure (people/policies and property) and will both remain, however, they may impact the coefficient stability. Additionally, the models utilized (random forest and XGBoost are less sensitive to multicollinearity reducing the impact of overall model performance).

The exploratory analysis performed provides preliminary support to the research question and hypothesis that tornado hazards in the United States may be gradually shifting towards the Southeast's Dixie Alley region over time. Figure 2 suggests that the annual tornado counts in Tornado Alley may have declined in the more recent years, while tornado counts within Dixie Alley has been gradually increasing. Figure 5 suggests that stronger tornadoes (F/EF3 and greater) have been recorded more frequently within the Dixie Alley region, specifically during the more recent years, although both regions show considerable variability over time. Tornado Alley also exhibits greater outliers with the exposure data for population and housing units. In Figure 6, the correlation matrix shows strong positive relationships between housing units and population estimates as well as the normalized measures for tornadoes per 10,000 housing units and 100,000 population.

4 Models

4.1 Preprocessing

Preprocessing techniques were implemented prior to predictive modeling. Highly right-skewed variables were log-transformed to reduce influence of extreme values, regional classifications were encoded as factor variables and normalized tornado frequency variables were created to account for differences in population and housing density across counties.

Principal Component Analysis (PCA) was explored as an unsupervised dimensionality reduction technique. PCA was conducted utilizing population estimate, housing unit estimate, median home value estimate, average tornado length/width, tornadoes per 100,000 population and tornadoes per 10,000 housing units.

The PCA results showed that the first three principal components explained approximately 83% of the total variance in the dataset, and the first four explained approximately 92.5%. These results suggest moderate multicollinearity between predictors. However, as the variables are all important for the insurance stakeholders, and random forest and XGBoost are less sensitive to multicollinearity, the variables will be retained instead of replacing them with principal components.

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	1.5879	1.3466	1.2149	0.81477	0.70416	0.1587	0.06560
Proportion of Variance	0.3602	0.2590	0.2109	0.09483	0.07084	0.0036	0.00061
Cumulative Proportion	0.3602	0.6192	0.8301	0.92495	0.99579	0.9994	1.00000

Figure 7: Proportion of variance by each principal component

4.2 Model Selection

Three supervised learning models were used to predict whether a county would be classified as a high tornado hazard next year: logistic regression, random forest and XGBoost. Logistic regression served as the baseline model as it supports the binary classification of the target variable. Additionally, random forest and XGBoost were utilized as flexible models to analyze if performance improves when compared to logistic regression.

4.3 Feature Importance

Feature importance was addressed for each model. The logistic regression model looked to coefficient magnitude with L1 Regularization (Lasso). Interpretations include that the region categories for Dixie Alley showed a strong positive relationship in addition to the tornado characteristics for average length and width. Housing unit estimate and region category for Tornado Alley had showed a positive relationship as well while median value estimate showed a negative relationship and population estimate dropped to zero.

Random forest utilized mean decrease gini importance. From these results in order of importance from highest to least include: average length, median value, population, housing unit, average width and region category.

Lastly, XGBoost utilized gain importance. The order of rank by gain from highest to lowest includes: average length, median value, population, average width, housing unit, and region category.

As each model determines importance with a different scale, to compare the feature importance across all three they were normalized and combined to plot the following as a visualization:

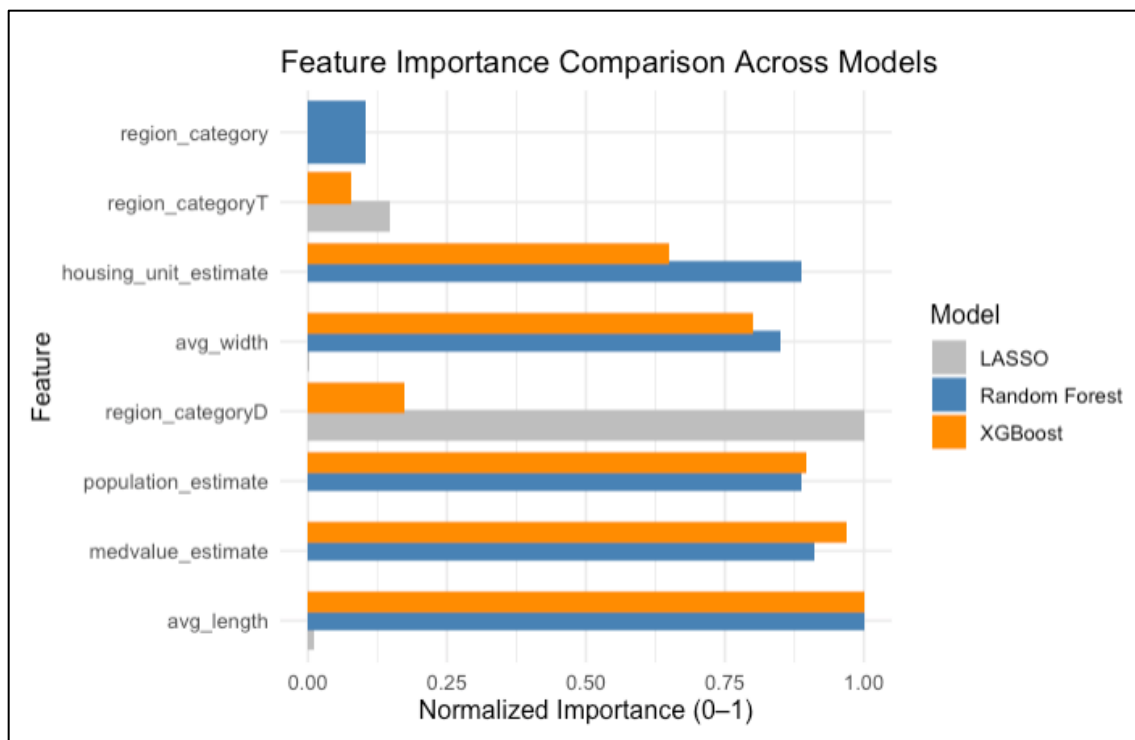


Figure 8: Feature Importance comparison across each model (logistic regression, random forest and XGBoost)

Random forest and XGBoost tended to align consistently on feature importance. From all models together, the most important predictor appears to be average length. Median value and population also appeared to be important predictors. Average width and housing unit appear to be moderately important while region is ranked highly in the logistic regression model, however, does not appear to be important in the random forest or XGBoost models. However, I will not be removing any variables due to the importance of retaining region category for the purposes of the research question.

4.4 Model Evaluation Methods, Assumptions & Overfitting

Model performance was evaluated using recall and Precision-Recall Area Under the Curve, PR-AUC. Recall was prioritized as the primary evaluation metric to ensure identification of high-risk counties. This is essential for the stakeholder as failure to accurately identify high-risk

counties could contribute to the continuation of underwriting in high insurance exposure areas causing an increase in financial risk. Confusion matrices also provided a visualization of model performance.

The dataset was split using a time-based approach for county-year observations. The training set included the earlier years of 1979-2014 and the testing set included the later years of 2015-2024. This helped to ensure the model was evaluated on future observations, aligning with the forecasting goal of this project.

To further assess the model, cross-validation included stratified k-fold ($k=5$) and repeated k-folds ($k=5$, repeats = 3) that were applied to this training set to also address overfitting, ensuring that observations from both Tornado Alley and Dixie Alley and high-risk and low-risk counties were represented in each fold.

The model assumptions for logistic regression includes linearity between predictors and the log odds of the dependent variable, independence and the absence of perfect multicollinearity while data is collected randomly (Model 5, Models & Assumptions) (Geeks for Geeks, 2025). As stated earlier, multicollinearity was addressed with a correlation matrix which did indicate correlation between some variables. Additionally, skewed variables for population and housing units were addressed by completing a log transformation. Property loss, which was also highly right-skewed, contributed to the composite hazard score for the target variable and was log transformed and standardized with z-scores.

Similarly to logistic regression, random forest and XGBoost also assume that the data was collected randomly and independently (Model 5, Models & Assumptions) (Geeks for Geeks, 2025).

To address assumptions for each model, multicollinearity was reviewed utilizing the correlation matrix. Though this did show multicollinearity was observed with some variables, such as population and housing, these were not deemed necessary to remove. Retaining these variables may lead to unstable coefficient estimates, however, due to their pertinent nature to this project to insurance exposure this is necessary. Additionally, multicollinearity does not significantly impact performance for models like random forest and XGBoost. Linearity was reviewed utilizing log transformations for highly right-skewed variables.

4.5 Hyperparameter Tuning

Hyperparameter tuning used randomized search with a 5-fold cross-validation to reduce the risk of overfitting. This was applied to the random forest and XGBoost models leaving the baseline Logistic Regression model untuned.

Tuning was conducted utilizing the training set only and `na.omit()` to remove any rows that may have had missing values so the models don't break. The train set was split previously with a time-based approach including the years 1979-2014 as an overfitting control method.

Random forest used the caret package for hyperparameter tuning (`method = rf`). This determined that the optimal number of variables to consider at each split resulted in a `mtry` of 2. A custom summary function was created to ensure the model was tuned using PR-AUC versus the default caret option ROC. The tune length was set at 5. This resulted in a model that was able to very accurately predict high-risk counties ($\text{Sens} = 0.922$), however, incorrectly flagged many low-risk counties as well ($\text{Spec} = 0.110$).

```

Random Forest

23595 samples
  6 predictor
  2 classes: 'NoRisk', 'HighRisk'

No pre-processing
Resampling: Cross-Validated (5 fold)
Summary of sample sizes: 4718, 4719, 4720, 4718, 4720
Resampling results across tuning parameters:

mtry  PR_AUC
2     0.3809107
3     0.3784014
4     0.3770345
5     0.3761925
7     0.3751180

PR_AUC was used to select the optimal model using the largest value.
The final value used for the model was mtry = 2.

```

Figure 9: Hyperparameter tuning results for random forest

XGBoost also used the caret package for hyperparameter tuning. The optimal model resulted in: `nrounds = 50`, `max_depth = 1`, `eta = 0.3`, `gamma = 0`, `colsample_bytree = 0.6`, `min_child_weight = 1` and `subsample = 1`. The maximum tree depth of 1 shows that shallow trees performed well and was able to capture tornado risk.

4.6 Model Training & Selection

Model performance was compared using recall and PR-AUC across both stratified and repeated k-fold cross-validation (figures 11 & 12). The below provides a chart of the results for each model comparing the results between the stratified k-fold cross-validation ($k=5$) and the repeated k-fold cross-validation ($k=5$, $\text{repeats} = 3$).

Model	Recall sensitivity	PR-AUC	Interpretation
Logistic Regression	94.20%	0.6861	Baseline model with no tuning. High recall. Overpredicts high-risk counties resulting in many false positives.

random forest	51.49%	0.6745	Misses too many high-risk counties.
XGBoost	90.05%	0.7043	Most balanced model. High recall and strong PR-AUC.

Table 3: Stratified k-fold cross-validation (k=5): Comparing recall and PR-AUC results

Model	Recall sensitivity	PR-AUC	Interpretation
Logistic Regression	90.29%	0.3951	Baseline model with no tuning. High recall but weaker precision recall.
random forest	43.78%	0.3924	Low recall.
XGBoost	78.82%	0.4089	Remains most balanced.

Table 4: Repeated k-fold cross-validation (k=5, repeats = 3): Comparing recall and PR-AUC results

Logistic Regression Prediction	Predicted No Risk	Predicted High Risk
Actual No Risk	180 (TN)	1716 (FP)
Actual High Risk	186 (FN)	3022 (TP)

Table 5: Logistic Regression Model Confusion Matrix Results

random forest Prediction	Predicted No Risk	Predicted High Risk
Actual No Risk	1106 (TN)	790 (FP)
Actual High Risk	1556 (FN)	1652 (TP)

Table 6: random forest Model Confusion Matrix Results

XGBoost Prediction	Predicted No Risk	Predicted High Risk
Actual No Risk	311 (TN)	1585 (FP)
Actual High Risk	319 (FN)	2889 (TP)

Table 7: XGBoost Model Confusion Matrix Results

Logistic regression achieved the highest recall sensitivity of the 3 models (94.20%). As shown in Table 3, it correctly identified a large number of high-risk counties (true positives), however, it also produced many false positives, which indicates that low-risk counties were incorrectly classified as high-risk.

Alternatively, random forest missed high-risk counties with a large false negative number (Table 3) and did not achieve an optimal balance with lower recall and poorer performance. As seen in Table 7, XGBoost provided the best balance with a high recall sensitivity of 90.05% and the highest PR-AUC at 0.7043 with stratified k-fold cross-validation. This is supported by the confusion matrix achieving a strong number of true positives and lower false positives than random forest with a better balance of false negatives. Having this balance is important for the stakeholder to accurately identify high-risk counties and minimizing missed risk.

Repeated k-fold cross-validation performed similarly for recall, however, PR-AUC drastically lowered for each model. This is due to repeated sampling being introduced which provides a more realistic model performance. It also has a higher run-time than the stratified k-fold cross-validation method. Cross-validation methods were compared to the baseline Logistic Regression model. Logistic Regression achieved a high recall, accurately identifying high-risk counties, however, it also produced a large amount of false positives (overpredicted high-risk counties) and random forest showed underfitting. XGBoost demonstrated the most balanced method between both cross-validation methods which reinforced the choice as the best performing model.

The model performance on the test set was consistent with cross-validation results. This suggests the model is not significantly overfit. Logistic regression had the highest recall, however, it suggested overpredicting for true positives. Logistic regression showed higher bias,

random forest reduced bias but exhibited greater variance and XGBoost provided the strongest balance between bias and variance

To improve sensitivity, the target variable hazard cutoff for binary classification was adjusted from the original 75th percentile to the 65th percentile allowing for the top 35% of county-years to be classified as high-risk. While this increases false positives, this adjustment helped to reduce the risk of missed high-risk observations and aligns with the overall goal to prioritize risk identification.

XGBoost was ultimately selected as the best-performing model for this project. It was the most balanced and maintained high recall while achieving a strong precision-recall across both validation strategies. The recall sensitivity was strong at 91.15% with the highest PR-AUC of 0.7075. This outperformed both Logistic Regression and random forest. This aligns with prior findings that as XGBoost has the ability to handle large-scale data and performs better than random forest with class imbalances (Geeks for Geeks, 2025).

4.6 Post Hoc Analysis

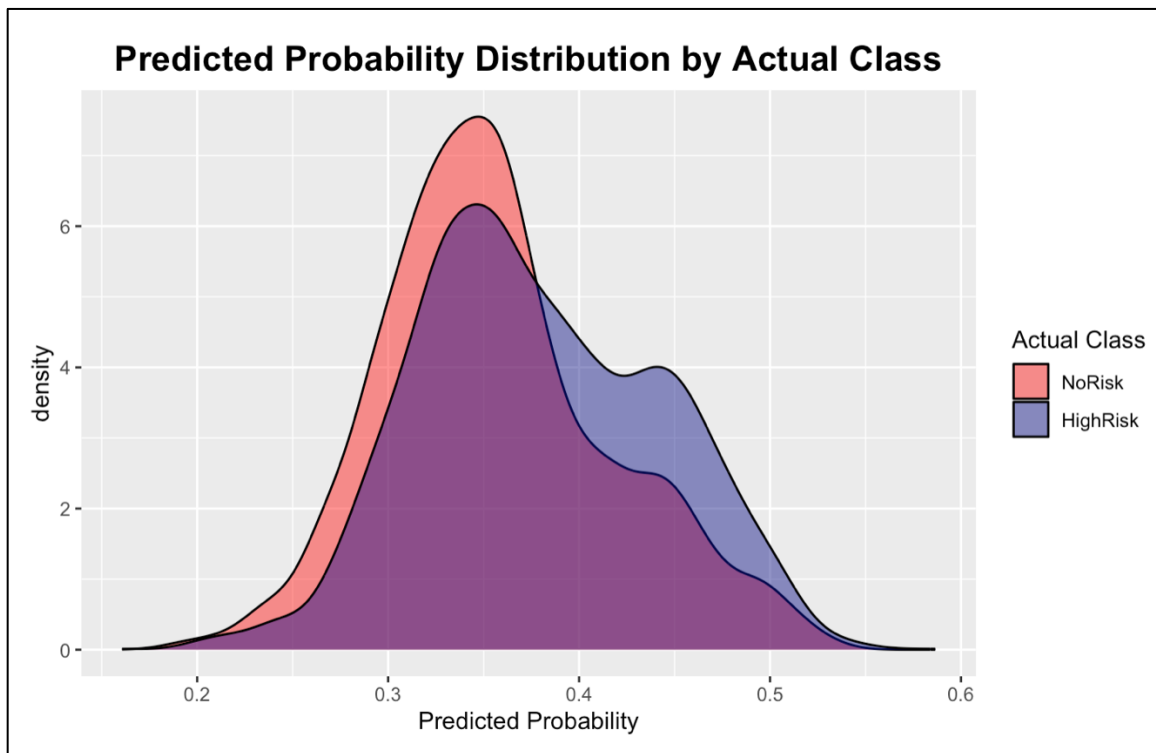


Figure 10: Ad Hoc Analysis of XGBoost model's predicted probability distributed by class (no risk vs. high risk)

Post hoc analysis provides additional insight into model performance and supports overall findings. Figure 10 shows distribution of predicted probability from the XGBoost model by class, either no-risk or high-risk. As shown, high-risk counties are generally shifted to the right indicating a higher score to counties that are true positives with a lower score for true negatives. However, there is substantial overlap indicating false positives and false negatives as well.

Table 8 shows the baseline distribution of counties by region category (T = Tornado Alley, D = Dixie Alley and O = Other), where Dixie Alley accounts for approximately 25% of the counties. Table 9 shows the distribution predicted by the XGBoost model for high-risk counties for next-year tornado hazard by region category. In this comparison, Dixie Alley now accounts for approximately 29% of the predicted high-risk counties compared to Tornado Alley at 22%.

This indicates that there is a larger amount of counties in Dixie Alley classified as high-risk next-year compared to the baseline, suggesting support for the original hypothesis that there is a shift towards this region.

Region	Number	Percent
Other	2616	51.25%
Tornado Alley	1189	23.30%
Dixie Alley	1299	25.45%

Table 8: Distribution of counties by region category (T = Tornado Alley, D = Dixie Alley, O = Other) in the model test set showing the baseline proportion of counties prior to prediction.

Region	Number	Percent
Other	2166	48.41%
Tornado Alley	1009	22.55%
Dixie Alley	1299	29.03%

Table 9: Distribution of predicted high-risk counties for next-year tornado hazard by region category. The proportion of counties classified as high-risk by the XGBoost model across Tornado Alley, Dixie Alley and Other.

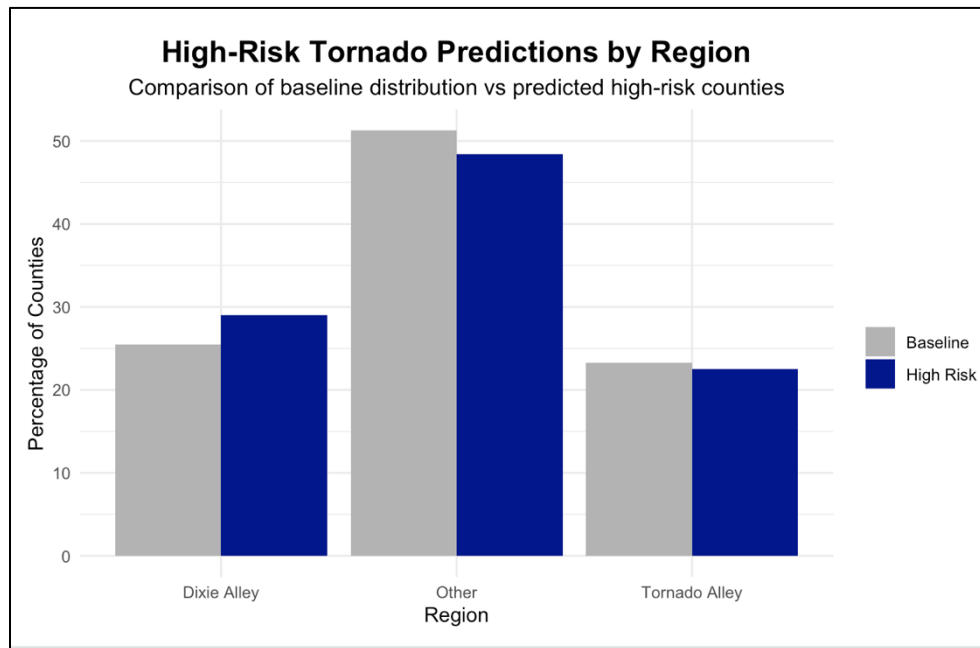


Figure 11: Baseline distribution of counties across regions compared to counties predicted as high-risk

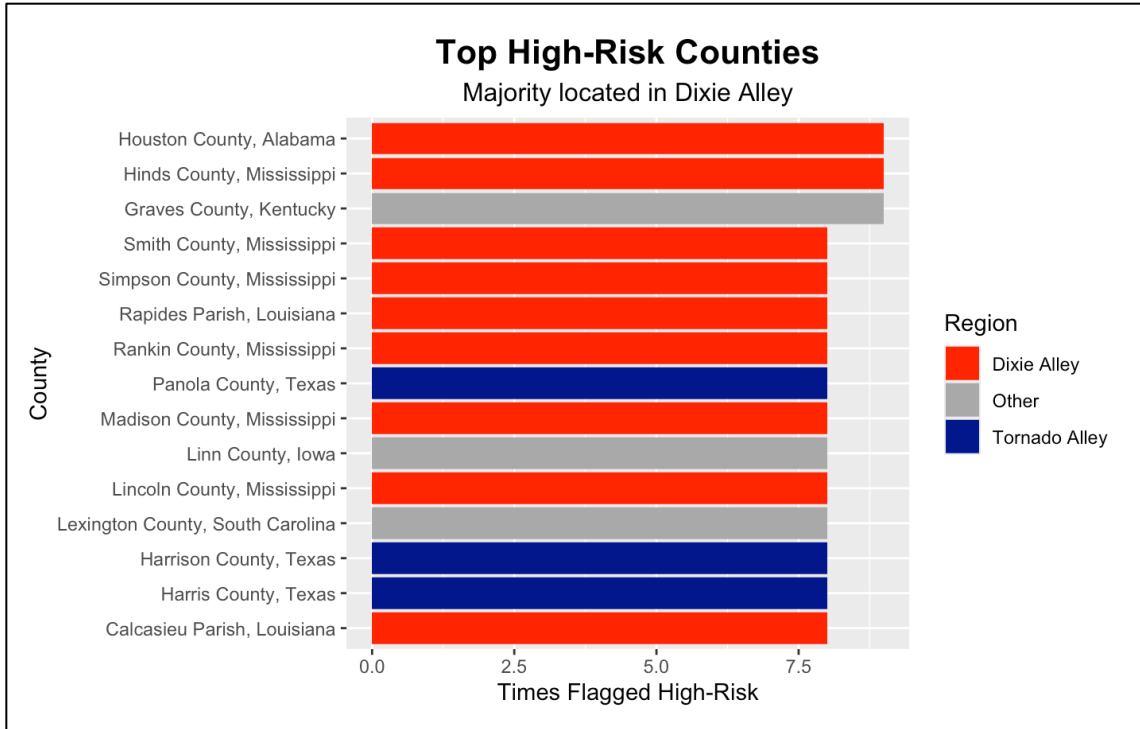


Figure 12: Top high-risk consistently identified (number of times flagged as high-risk) by region

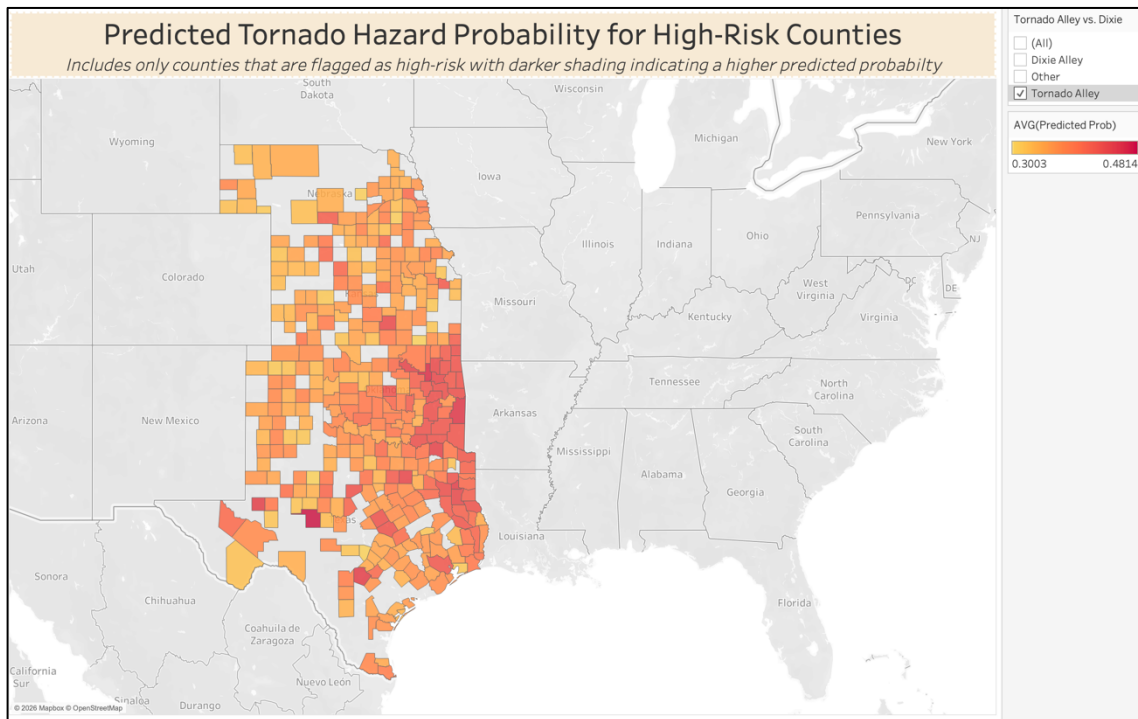


Figure 13: Tableau Dashboard visualization showing predicted tornado hazard probability for high-risk counties within Tornado Alley. Includes only counties that flagged as high-risk from the XGBoost model with darker shading indicating higher predicted probability

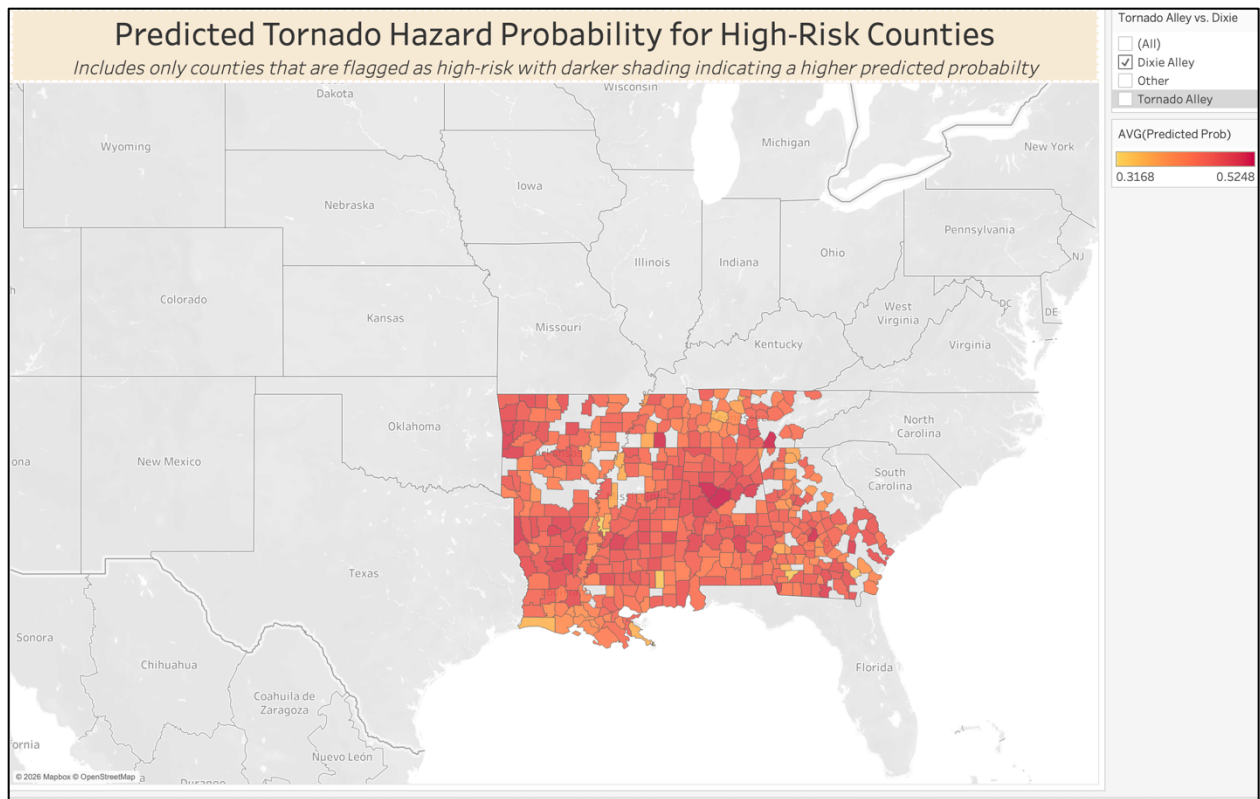


Figure 14: Tableau Dashboard visualization showing predicted tornado hazard probability for high-risk counties within Dixie Alley. Includes only counties that flagged as high-risk from the XGBoost model with darker shading indicating higher predicted probability

Figures 11-14 further support the hypothesis and prediction suggests an increasing concentration of high-risk tornado probability in the Dixie Alley region counties. Most counties that continuously flagged as high-risk occurred within this region with Mississippi occurring most often. Further analysis also shows that counties within Dixie Alley have a slightly lower average population and home value compared to Tornado Alley. However, combined with the model results showing more high-risk predictions within this region, it does suggest that tornado hazard is increasingly occurring in areas with meaningful insurance exposure.

To continue analysis into insurance exposure, Table 10 below shows the counties classified as high-exposure (top 25% of median home value) per region. This shows that a larger number of the top 25% of median home value homes are located within the Dixie Alley region. Though

they do not have the highest average home value (Table 10), this is still a meaningful concentration with elevated insurance exposure.

Together, the results suggest that tornado hazard is shifting geographically towards Dixie Alley, however, it is also occurring in areas that show meaningful insurance exposure, which reinforces the need for potential risk analysis and updated underwriting strategies in this region.

Region	Average Population	Average Home Value
Other	171,065	218,289
Tornado Alley	151,036	176,371
Dixie Alley	80,331	170,448

Table 10: The average population estimate and average home value between each region (T = Tornado Alley, D = Dixie Alley and O = Other)

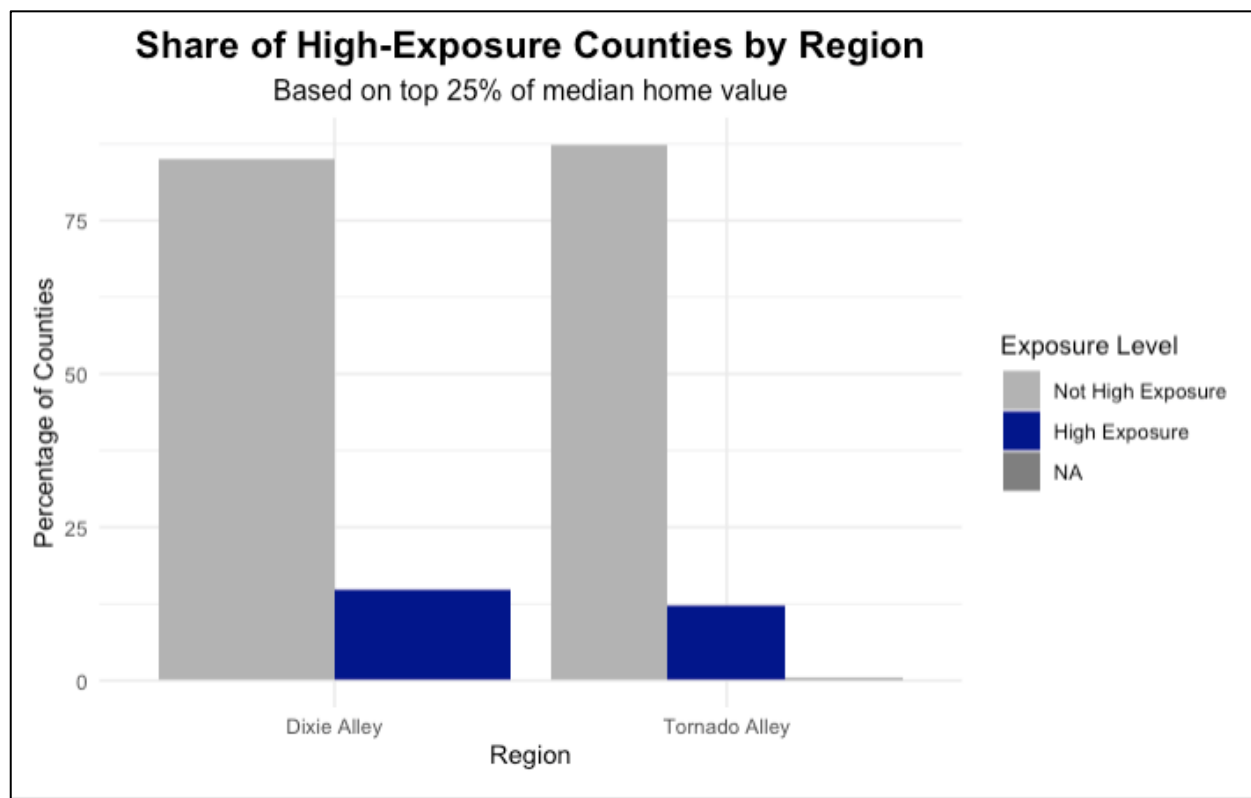


Figure 15: Distribution of counties that are classified as high insurance exposure (top 25% of median home value) by region

5 Discussion & Next Steps

The primary objective of this project was to determine whether tornado hazard in the United States is shifting geographically from Tornado Alley towards Dixie Alley, and whether counties with elevated tornado hazard also exhibit increased insurance exposure. The results provided evidence consistent with the hypothesis. The exploratory data analysis and model predictions indicated there was an increasing concentration of tornado hazard within Dixie Alley. Table 9 showed that Dixie Alley accounted for a higher proportion of high-risk counties compared to the baseline distribution in Table 8.

Among the three models evaluated, XGBoost performed the best achieving a strong balance between high-risk counties and limited missed risk. XGBoost achieved a recall sensitivity of approximately 91% while also producing the strongest PR-AUC performance amongst all models. The model effectively captured most high-risk observations aligning with the stakeholder's goal.

The findings supporting a tornado hazard shift from Tornado Alley towards Dixie Alley also supported research by Gensini and Brooks (2018). The project extended this research with an insurance lens to predict if shifting tornado exposure also aligned with higher population and housing exposure. However, Dixie Alley did show a lower average population and median home value compared to other regions (Table 10). However, Dixie Alley contains a higher number of high exposure homes, with a greater number valued within the top 25% of median home value by region alerting to a meaningful insurance exposure.

From a stakeholder perspective, this suggests that insurance companies may want to adjust underwriting and pricing strategies for this region because increasing tornado hazard

concentration in Dixie Alley may contribute to greater claim frequency and financial exposure. Adjusting pricing strategies will also help with evolving hazard patterns.

Limitations should be considered with this analysis. The assumption of independence of observations, however, this is likely violated due to the spatial and temporal dependencies. As the data contains strong geographic structure, there is a risk that the model may learn spatial dependence across counties rather than relying on the intended predictor variables. There are implications that could impact the model's learning or performance by ignoring the spatial structure. Spatial dependency refers to observations in nearby locations being more similar and influenced by each other than those further apart (Noordam 2025). The analysis assumes independence of observations, however, this is likely to be violated due to the spatial and temporal dependencies. Ignoring this could lead to a potentially overly confident model performance with nearby counties sharing similar tornado risk characteristics leading to clustering patterns rather than a true independent relationship. These limitations represent potential threats to validity and missing data assumptions.

Missing, or unknown, data for tornado intensity and property loss was handled by replaced with NA then removing for modeling which assumes missingness is random and may have introduced bias.

Insurance exposure data came from population, median housing value and units data rather than from insurance loss data. Future work could also look at integrating actual insurance loss data to better capture insurance exposure.

Other variables not considered includes climate change trends and assumes that historical tornado patterns are predictive of future behavior.

Future work could address spatial lagging of predictors to better account for geographic relationships and improve generalizability. Additionally, integrating actual insurance loss data to better capture insurance exposure.

Overall, this project provided strong evidence that tornado hazard is shifting towards Dixie Alley and it includes meaningful insurance exposure. The XGBoost model supported this by identifying high-risk counties with a strong predictive performance. This highlights the importance of adjusting risk management and underwriting strategies for newly evolving hazard.

6 GitHub

<https://github.com/jill-baker/tornado-insurance-risk-repo>

7 Tableau Dashboard

https://public.tableau.com/app/profile/jill.baker1863/viz/Capstone_17772439582620/Sheet1?publish=yes

8 Appendix

The Data Dictionary below includes identification of key raw, engineered and target variables in the final county-year tornado hazard and insurance exposure dataset. The target variable, `high_tornado_hazard_next_year`, was created from the hazard-related variables in the dataset: `tornado_count_county_year`, `average_intensity_county`, `max_intensity_county`, `total_property_loss_year` and `strong_tornado_count`.

Column Name	Example	Data Type	Description
om	'192'	Integer	Tornado number. Count of tornadoes during given year from NOAA dataset
yr	'1950'	Integer	Year of tornado event (1950-2024)
mo	'3'	Integer	Month of tornado event, (1-12)

dy	'26'	Integer	Day of tornado event, (1-31)
date	'1950-03-26'	Date	Date of tornado event in yyyy-mm-dd format
st	'AR'	String	State (two-letter postal abbreviation) of tornado event
mag	'2'	Integer	F-Scale (EF-Scale after 01/07) Values 0, 1, 2, 3, 4, 5, -9 (-9=unknown)
loss	'4'	Numeric	Estimated property loss in millions of dollars. 0 or blank = unknown, not \$0
len	'17.4'	Numeric	Length of tornado in miles
wid	'150'	Integer	Width of tornado in yards
county_fips	'05019'	String	Federal Information Processing Standards code for county, full 5 digit code (first 2 digits for state, last 3 digits for county).
county_name_state	'Clark County, Arkansas'	String	Full county name and state
population_estimate	'21125'	Integer	Predictor. Present-day exposure variable. Estimated total population in county
housing_unit_estimate	'10029'	Integer	Predictor. Present-day exposure variable. Estimated total number of housing units in the county
medvalue_estimate	'155000'	Numeric	Predictor. Present-day exposure variable. Estimated median home value in the county (in dollars)
region_category	'T'	String	Predictor. Indicates if county is located in Dixie Alley, Tornado Alley or Other. Dixie Alley = D, Tornado Alley = T, Other = O.
high_tornado_hazard_next_year	'1'	Binary	Target variable. Indicates whether a county is classified as high tornado hazard in the following year based on county's forward-shifted composite hazard classification (1=high tornado hazard, 0=not high tornado hazard)
tornado_count_county_year	'4'	Integer	<i>Target creation hazard variable.</i> Total number of tornadoes per county for a given year
average_intensity_county	'3'	Numeric	<i>Target creation hazard variable.</i> Average tornado intensity (F/EF-Scale) for tornadoes in county for a given year.
max_intensity_county	'5'	Integer	<i>Target creation hazard variable.</i> The maximum/highest observed tornado intensity per county in given year
total_property_loss_year	'5'	Numeric	<i>Target creation hazard variable.</i> The total property loss with all occurred tornadoes per county in given year

average_property_loss	'4'	Numeric	Hazard variable. The average property loss for tornadoes in county for a given year
strong_tornado_count	'3'	Integer	<i>Target creation hazard variable.</i> Count of tornadoes in county-year where the tornado intensity (mag) is ≥ 3
property_loss_per_tornado	'50000'	Numeric	Calculation of total <code>property_loss_year</code> / <code>tornado_count_county_year</code>
tornadoes_per_100k_population	'100'	Numeric	Normalized measure: Calculation of <code>tornado_count_county_year</code> / <code>population_estimate</code> * 100000
tornadoes_per_10k_housing_units	'50'	Numeric	Normalized measure: Calculation of <code>tornado_count_county_year</code> / <code>housing_unit_estimate</code> * 10000
avg_length	'5'	Numeric	Predictor. Average length in miles for tornadoes per county
avg_width	'5'	Numeric	Predictor. Average width in yards for tornadoes per county
composite_hazard_score	'-0.50'	Numeric	Composite index created by averaging standardized tornado frequency, intensity, and loss variables. Used to classify high hazard counties.
high_tornado_hazard	'1'	Binary	Variable used to create the predictive target <code>high_tornado_hazard_next_year</code> and represents the current-year classification
log_property_loss	'1'	Numeric	Logged transformation of <code>total_property_loss_year</code>
log_population	'1'	Numeric	Logged transformation of <code>population_estimate</code>
log_housing	'1'	Numeric	Logged transformation of <code>housing_unit_estimate</code>
z_avg_intensity	'1'	Numeric	Z-Score from <code>average_intensity_county</code> for target variable creation: <code>high_tornado_hazard_next_year</code>
z_max_intensity	'1'	Numeric	Z-Score from <code>max_intensity_county</code> for target variable creation: <code>high_tornado_hazard_next_year</code>
z_strong_count	'1'	Numeric	Z-Score from <code>strong_tornado_count</code> for target variable creation: <code>high_tornado_hazard_next_year</code>
z_property_loss	'1'	Numeric	Z-Score from <code>log_property_loss</code> for target variable creation: <code>high_tornado_hazard_next_year</code>
z_tornado_count	'1'	Numeric	Z-Score from <code>tornado_count_county_year</code> for target variable creation: <code>high_tornado_hazard_next_year</code>

strong_tornado_fraction	'1'	Numeric	Calculation of strong tornado count divided by total tornado count per county year
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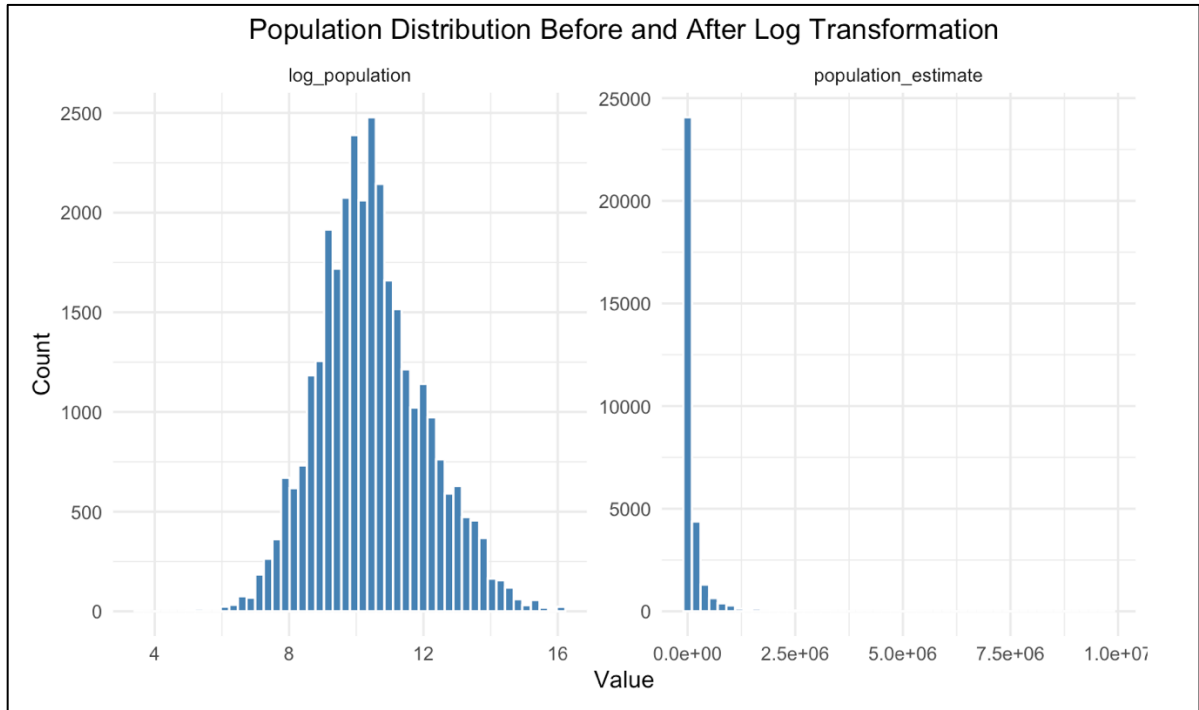


Figure 16: Histogram comparing population distribution before and after log transformation.

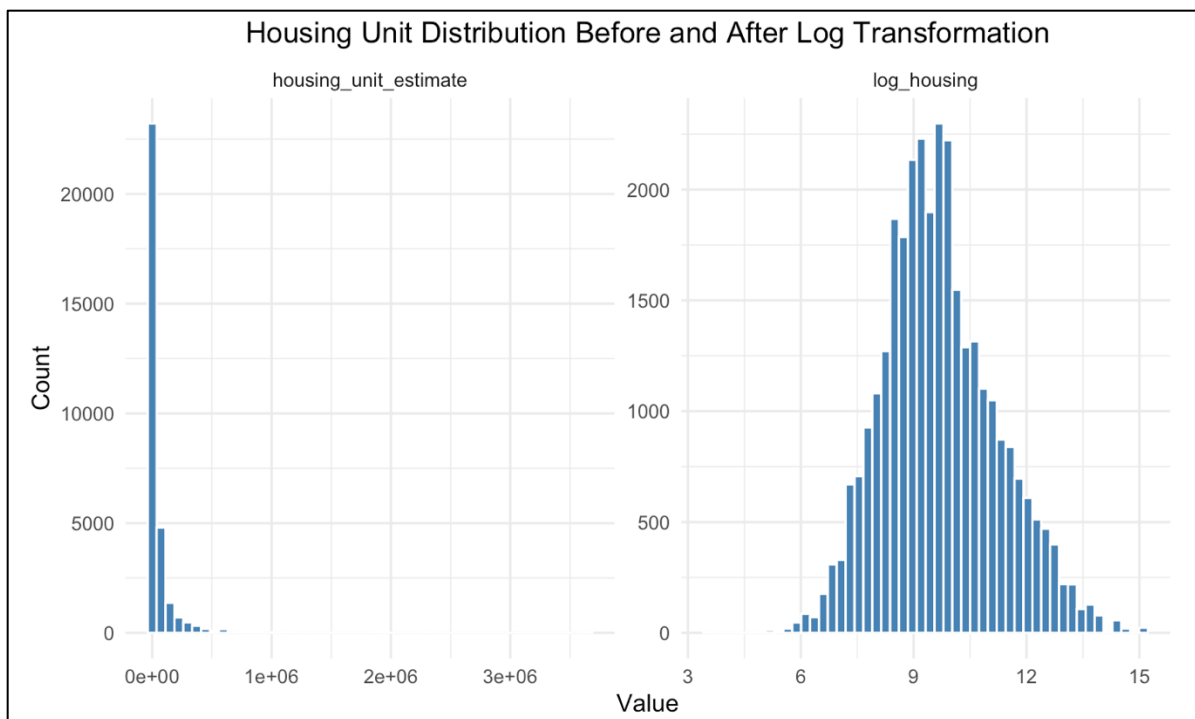


Figure 17: Histogram comparing housing unit distribution before and after log transformation.

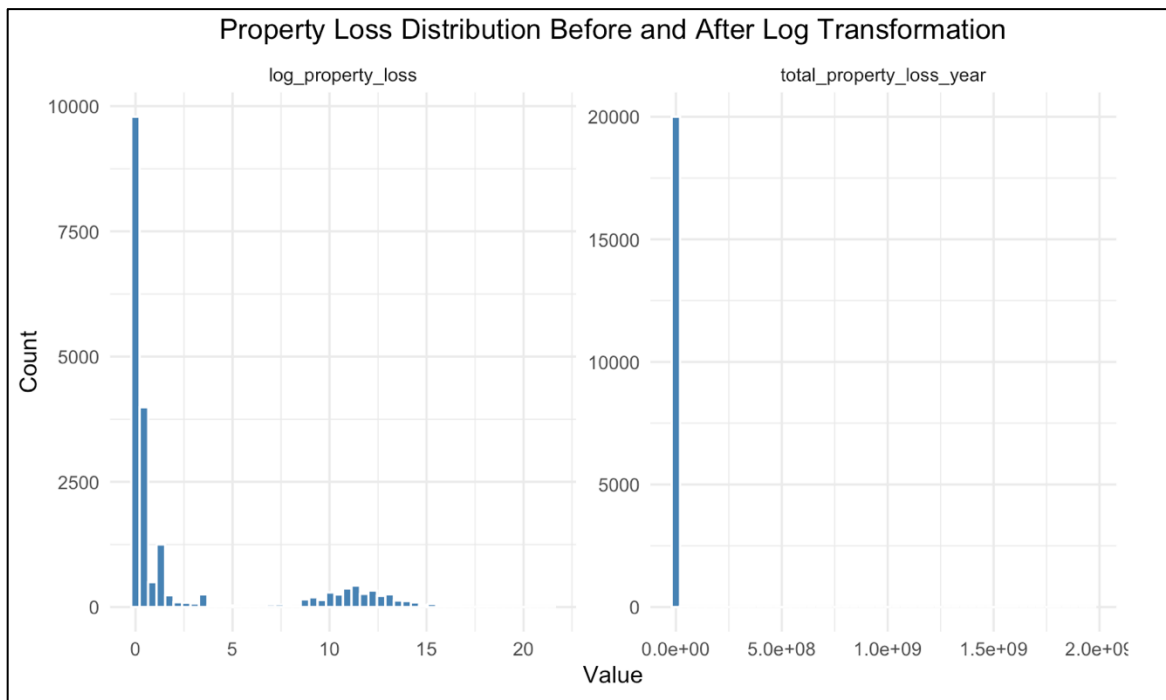


Figure 18: Histogram comparing property loss distribution before and after log transformation.

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